

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 1.1: Taxonomy I — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 1.1: Taxonomy I
Driving Question	DRIVING QUESTION: How can we organise the many different living organisms around us so that everyone can identify and study them in the same way?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Introduction to Taxonomy and Classification: Why Do Our Groupings Differ?	Teacher note: Learners sorted a set of Kenyan organism cards (maize, bean plant, sukuma wiki, tilapia from Lake Victoria, cow, chicken, mushroom, acacia, housefly) into groups of their own choosing. Expect groupings by colour, habitat, edibility or usefulness (e.g. 'foods we eat', 'animals on the farm'). Different groups produced different groupings for the same organisms, and some organisms (e.g. tomato: fruit or vegetable?) did not fit neatly. Use WAIT TIME of 5 seconds after asking 'Why did your group put these together?' and cold-call 3 groups to share their criteria.	Organisms can be grouped in many ways depending on the criteria chosen. Taxonomy is the science of naming, describing and classifying organisms. Local names vary between communities (e.g. the same plant can have different names in Kikuyu, Luo and Kiswahili), which causes confusion. Scientists therefore need agreed, shared criteria so that everyone groups organisms the same way. Pedagogical note: do not correct learners' first groupings; use the differences to create the need for a common system.	Answer key: Groupings differed because each group chose its own criteria (colour, use, habitat), and none was 'wrong' but none was universal. To organise organisms so everyone can identify and study them the same way, we need standard, observable shared characteristics and agreed names, which is what taxonomy provides. Add to the Driving Question Board: 'What shared criteria do scientists use?' and 'Why do local names cause confusion?' Model v1: a simple sketch showing the same organisms in two different groupings, with a note that a common system is needed.
Lesson 2 Importance of Classification: Why Do We Need to Identify Organisms Correctly?	Teacher note: Learners worked through Kenyan case scenarios, e.g. a farmer in Kericho misidentifying a pest on tea, a health worker confusing a harmless snake with a venomous one, and conservation officers in Maasai Mara or Tsavo counting and protecting species. Learners noted the problems caused by wrong identification: wrong pesticide used, crops lost, people harmed, endangered species overlooked.	Classification is important because it helps us to identify organisms correctly, study them systematically, communicate about them clearly, and conserve biodiversity by knowing what species exist and which are threatened. It also helps predict characteristics of related organisms (e.g. related crops share pests and diseases) and helps us find relationships between organisms. Pedagogical note: connect each	Answer key: We need to identify organisms correctly to avoid harm and wasted resources (wrong pest control, dangerous snake misidentification), to share knowledge accurately between farmers, doctors and scientists, and to protect rare species such as the mountain bongo. A common classification system makes this possible for everyone. DQB additions: 'How are groups arranged from large to small?'

	Ask 'What went wrong in this case, and what would have helped?' and cold-call 3 learners after 5 seconds of WAIT TIME.	benefit to a Kenyan example so learners see relevance beyond the textbook.	Model v2: add a box to the model labelled 'Purpose: identify, study, communicate, conserve' with one Kenyan example.
Lesson 3 Taxonomic Units: From Kingdom to Species	Teacher note: Learners examined a set of nested boxes or concentric circles and a classification chart for familiar Kenyan organisms (e.g. maize, domestic cat, human, tilapia) from kingdom down to species. They observed that each rank sits inside the one above and that organisms in the same lower rank are more alike. Use a mnemonic (e.g. 'King Philip Came Over For Good Soup') and invite learners to create a Kenyan one. Ask 'Which rank holds the most organisms? Which the fewest?' with 5 seconds WAIT TIME, then cold-call 3 learners.	The taxonomic ranks are nested: each rank sits inside the one above it. Kingdom is the broadest group and Species is the most specific. Every organism can be placed in the sequence kingdom, phylum, class, order, family, genus, species. Species is a group of organisms that can interbreed and produce fertile offspring. Pedagogical note: common errors include reversing the order or treating ranks as separate lists rather than nested; use the nested-box model to correct this.	Answer key: Taxonomic units are the ranks used to place an organism in progressively more specific groups, so any scientist can locate it in the same position. This helps organise the huge diversity of organisms in an orderly, shared way. DQB additions: 'How do the ranks make a hierarchy?' 'How do we name the species?' Model v3: add the seven ranks as nested boxes with one worked Kenyan example, such as maize or the domestic cat.
Lesson 4 Hierarchy of Classification: Climbing the Taxonomic Staircase	Teacher note: Learners built a 'staircase' or pyramid by arranging rank cards from kingdom down to species and placing example organisms on it, e.g. comparing the lion and the domestic cat, or maize and wheat, to see at which step they separate. They counted how the number of organisms decreases and the similarity increases at each step down. Ask 'At which rank do these two organisms first differ?' with 5 seconds WAIT TIME; cold-call 3 pairs.	Taxonomic groups form a hierarchy from broad to specific: each step down has fewer organisms that are more similar to each other. Species at the bottom share the most characteristics, while kingdoms share only a few basic ones. Closely related organisms share more ranks, e.g. lion and leopard share the same genus (Panthera), while lion and domestic cat share the same family only. Pedagogical note: stress that more shared ranks means closer relationship.	Answer key: Moving down the hierarchy narrows the group and increases similarity, so each organism can be placed in one precise position and compared with others. This makes it possible for everyone to identify and study organisms in the same way. DQB additions: 'What should each species be called, and how?' Model v4: revise the model into a staircase showing decreasing numbers and increasing similarity, with two Kenyan organisms traced down to where they split.
Lesson 5 Scientific Names and Binomial Nomenclature: One Name for Every Organism	Teacher note: Learners compared local names of the same organism in different communities and languages (e.g. maize, tilapia, sukuma wiki or kale) and then looked at their scientific names such as Zea mays, Oreochromis niloticus and Brassica oleracea. They noticed that scientific names have two parts and are written in a special style. Ask 'What do you notice about how these names are written?' with 5 seconds WAIT TIME; cold-call 3 learners.	Scientific names give every organism one universal name that scientists everywhere can use. A scientific name has two parts, the genus (capital letter first) and the species (all lowercase). Names are written in italics, or underlined when handwritten, and are usually from Latin or Latinised words. This system is called binomial nomenclature and was developed by Carl Linnaeus. Pedagogical note: learners often capitalise both words; model the rule repeatedly.	Answer key: A universal two-part name removes the confusion caused by local names, because Zea mays means maize to scientists in Kenya, Brazil or China alike. This answers part of the driving question by letting everyone refer to the same organism using the same name. DQB additions: 'How do we write and use scientific names correctly?' Model v5: add the genus and species name to the example organisms in the model, written with correct format.
Lesson 6 Applying Binomial Nomenclature: Naming	Teacher note: Learners practised writing scientific names for local organisms (e.g. Zea mays, Phaseolus vulgaris, Homo	We learned the principles of binomial nomenclature: two names (genus and species), correct capitalisation (genus	Answer key: Applying the rules consistently means that a student in Nairobi, a farmer in Kisumu and a scientist abroad all use the

Our Local Organisms the Same Way Everywhere	sapiens, Panthera leo, Acacia xanthophloea) and corrected a set of incorrectly written names (e.g. 'zea Mays', Panthera Leo, no underline). They then named organisms from the school compound using a simple key or reference list. Ask 'What is wrong with this name and how do we fix it?' with 5 seconds WAIT TIME; cold-call 3 learners.	capital, species lowercase), italics or underlining, and the use of Latin so names are international. Each species has only one accepted scientific name, and the genus can be abbreviated after first use (e.g. P. leo). Pedagogical note: check handwriting for underlining of each word separately.	same name for the same organism, so information can be shared and compared accurately. DQB additions: 'What characteristics are used to place organisms in the large groups, the kingdoms?' Model v6: add a labelled example showing a correctly written scientific name, with a short list of the writing rules.
Lesson 7 Five Kingdoms of Living Organisms	Teacher note: Learners examined specimens, images or microscope slides of Kenyan organisms from each kingdom: bacteria (e.g. from yoghurt or mursik), Amoeba or Paramecium from pond water, mushrooms or bread mould, maize or beans, and cattle or tilapia. They recorded features such as cell type, cell wall, number of cells and mode of nutrition. Ask 'Which features are the same within a kingdom, and which differ between kingdoms?' with 5 seconds WAIT TIME; cold-call 3 learners.	Scientists use fundamental characteristics such as cell structure (prokaryotic or eukaryotic), number of cells (unicellular or multicellular), presence of a cell wall and mode of nutrition to group organisms into the five kingdoms: Monera (Prokaryotae), Protocista, Fungi, Plantae and Animalia. These are different from colour, habitat or usefulness, which can vary within a kingdom and so are unreliable. Pedagogical note: a comparison table is a helpful tool; correct the misconception that fungi are plants.	Answer key: The five kingdoms are based on shared structural and nutritional characteristics that all members possess, so they give consistent groupings anyone can check, unlike our first groupings by colour or use in Lesson 1. DQB additions: 'How does a given organism get placed into a kingdom?' Model v7: add the five kingdoms with their key characteristics and one Kenyan example for each.
Lesson 8 Classification into the Five Kingdoms and Importance of Taxonomy: Our Final Explanation	Teacher note: Learners were given unfamiliar Kenyan organisms (e.g. a mushroom, a Spirogyra or algae sample, a bacterium from yoghurt, sukuma wiki, a chameleon) and used a simple table of characteristics to place each into a kingdom, then traced one organism through the full hierarchy and gave its scientific name. They revisited their Lesson 1 groupings and compared. Ask 'What evidence places this organism in this kingdom?' with 5 seconds WAIT TIME; cold-call 3 learners.	We learned that classification groups organisms using shared characteristics in a hierarchy (kingdom, phylum, class, order, family, genus, species); that each organism has a universal two-part scientific name; and that organisms are placed into one of five kingdoms based on cell type, body organisation and nutrition. Taxonomy is important for identifying, studying and communicating about organisms, and for conserving biodiversity. Pedagogical note: use this lesson to assess understanding against the first-lesson misconceptions.	Answer key: We can organise the many organisms around us by using agreed, observable shared characteristics to place each one in a nested hierarchy, and by giving each a universal binomial name. This allows everyone, anywhere, to identify and study organisms in the same way, which answers the driving question. Final DQB review: check off answered questions and note any that lead to Taxonomy II. Final model: learners present a complete model combining purpose, hierarchy, binomial naming and the five kingdoms, using a Kenyan organism as the worked example.

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